

A Smart Street Lighting Adaptive System

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Abstract

This poster reports the results of a research project concerned with the design, development and demonstration of an urban smart lighting system capable to autonomously control the street lamp lighting levels by exploiting real-time traffic data (number and classification of vehicles such as bus, car, motorcycle, bike and pedestrians) and weather conditions. The system is finalized to minimize the energy of the systems by selecting the minimum lighting level of each lamp complying with the legislative limits in all circumstances at night-time. The proposed light control strategy, which can be performed either on each single street lamp or on a group of them, has been demonstrated in two experimentation sites to allow energy savings up to 65% with respect to traditional street lighting systems.

INTRODUCTION

A recent survey of the European Commission pointed out that up to 80% of produced energy is used to supply cities power grids. **Smart Lighting** is a multidisciplinary area within the context of Smart City that provides integration among sensors, IoT and control technologies, with the aim of making efficient the street lamps management by reducing energy consumptions and light pollution, so as to save money for the community and improving the general well-being.

We present a smart lighting system prototype capable to autonomously control the street lamp lighting levels by exploiting traffic and weather data. The system is based on the use of local sensors, local controllers, IoT technologies, video-processing algorithms used for vehicles classification and motion detection, and remote web application.

SMART STREET LIGHTING SYSTEM

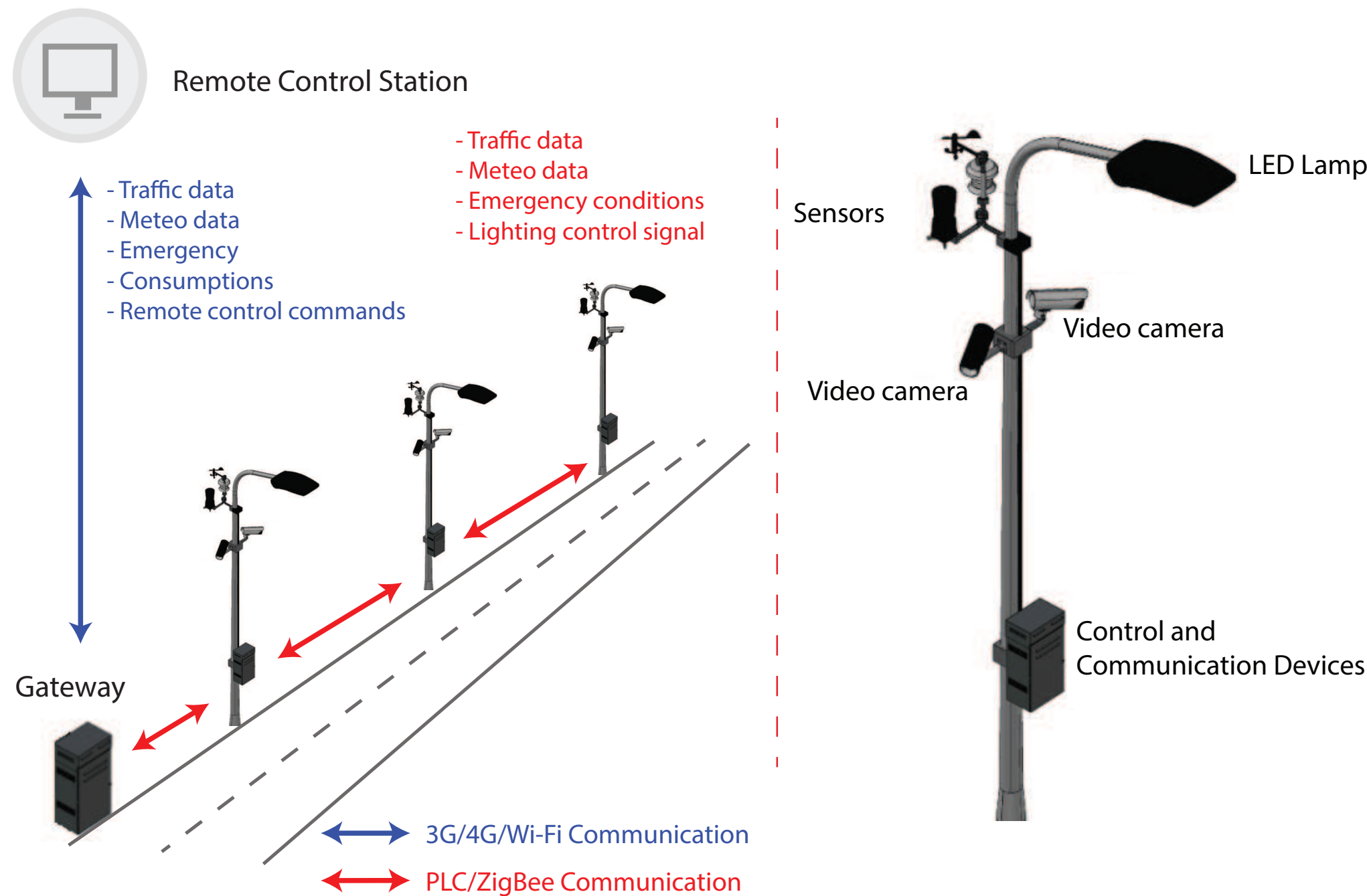


Figure 1: Smart Street Lighting Concept (left) and Smart Pole (right)

Main Features

1. Individual or group street light control;
2. Local communication between electronic devices based on Power Line Communication (PLC) and/or Zigbee devices;
3. Traffic, weather and emergency conditions (e.g. road accidents) monitoring;
4. Consumptions evaluation;
5. Remote control and management through web application;
6. Remote communication, between local electronic devices and web application, based on 3G/4G connection.

Working Mode

- **Day Time (Traffic Monitoring) Working Mode:** all traffic data are extracted in real-time through video processing and then shared with the web application.
- **Night Time Working Mode:** all smart functionalities are enabled and the LED light dimming is performed on the basis of vehicles or pedestrians presence in the monitored area (street, square, etc...). The LED light intensity is considered and managed as a function of vehicles/pedestrians presence and, in turn, a huge energy saving can be achieved whenever the traffic is null or low for long periods at night time.

Lighting Control

The video processing procedure is instrumental for lighting control. If vehicles or pedestrians are detected a "motion detected" event is generated and sent via local communication channel to the other poles

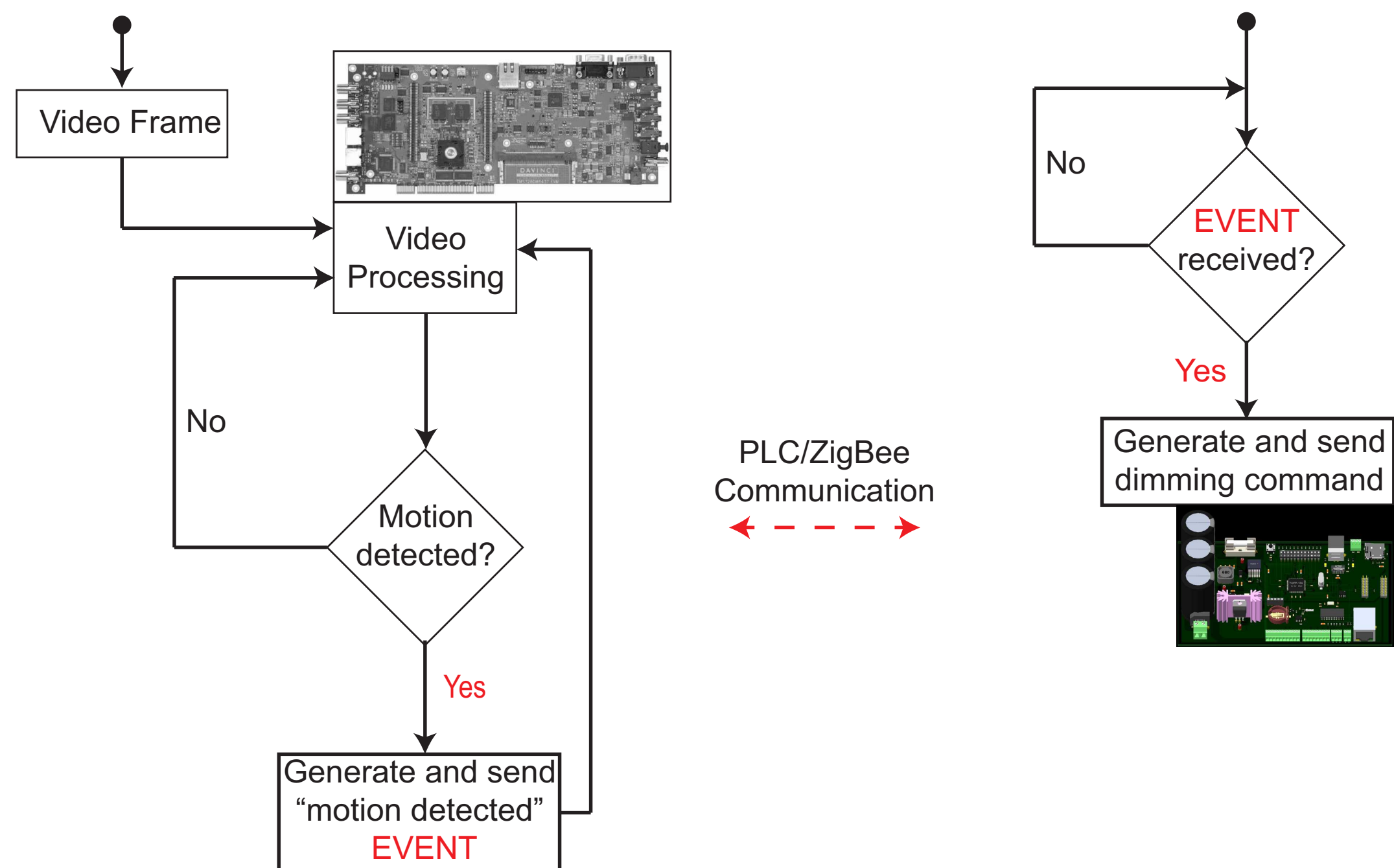


Figure 2: Video processing and lighting control boards interaction

RESULTS

Two experimental setups have been constructed in two different sites.

- **Experimentation site n.1:** busy road of University of Calabria (Italy)
- **Experimentation site n.2:** secondary road of the municipal district of Crucoli (Italy)

In both sites the experimental setup involves ten lighting poles deployed along a 150 [m] road.

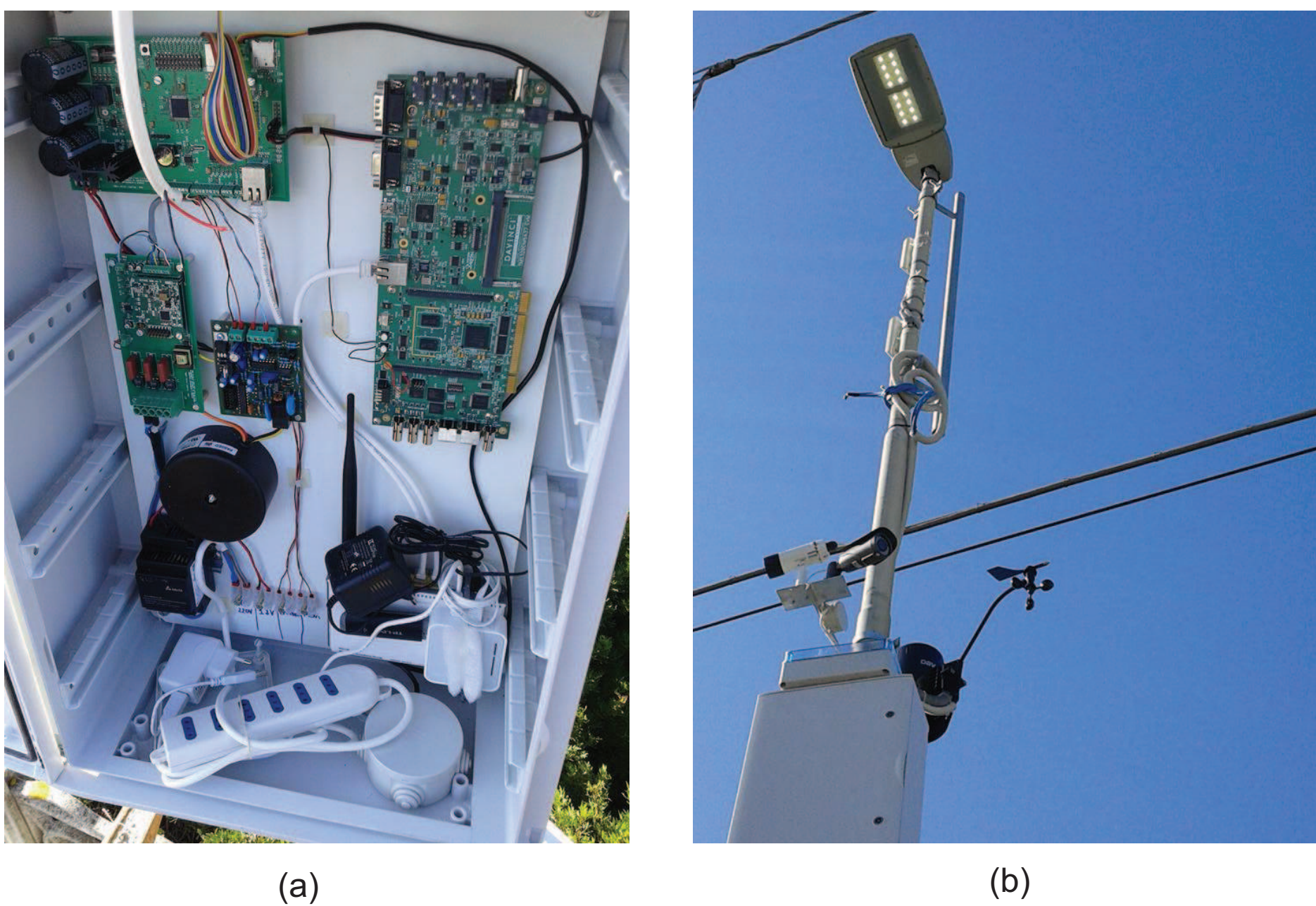


Figure 3: Setup of smart lighting pole.

The experimentation work (started in October 2015 and still ongoing) has been carried out in three steps.

1. Collecting a huge amount of data related to consumption of the traditional lighting systems already in order in the two demonstration sites;
2. Upgrading the ten street lamps to LED technology in both sites and collecting again the consumption data;
3. Replacement of the ten traditional street poles with ten smart poles in both the demonstration sites and collecting again the consumption data.

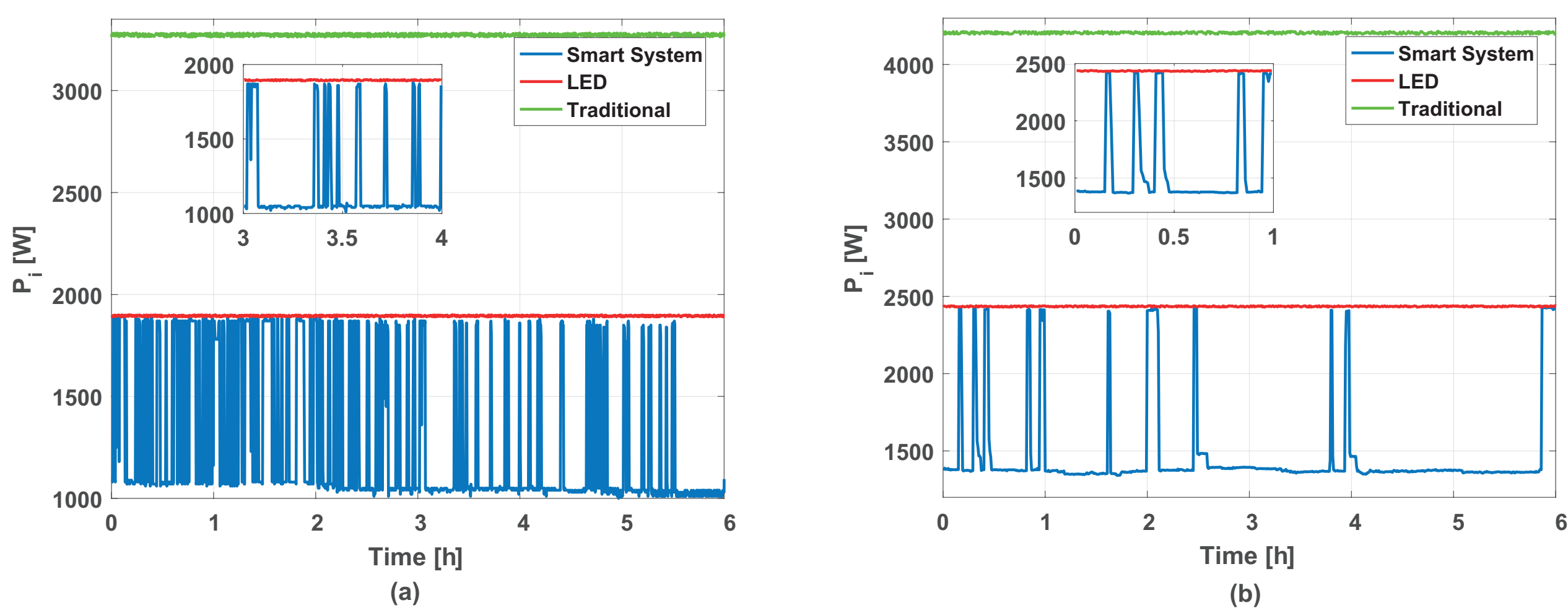


Figure 4: Consumptions evaluation for Site #1 (a) and Site #2 (b).

A twelve hours (from 18:00h to 06:00h) comparison in terms of consumption and energy saving is reported in Tables 1 and 2 respectively.

Site	Standard lighting	LED lighting	SMART lighting
#1	19.602,53 [Wh]	11.355,7 [Wh]	7.815,5 [Wh]
#2	13.143,23 [Wh]	7.614,52 [Wh]	4.649,7 [Wh]

Table 1: Consumptions

It is evident that our smart lighting prototype is capable to increase the energy saving up to 64,62% (while the substitution of traditional lamps with LED technology allows savings up to 42,07%) of energy when compared to the traditional system and up to 38,93% with respect to the LED upgraded system.

Site	SMART VS Standard	SMART VS LED	LED VS Standard
#1	60,13%	31,18%	42,07%
#2	64,62%	38,93%	42,07%

Table 2: Energy saving

Conclusions

- A smart street lighting prototype has been described and shown to be effective in energy saving in two experimental sites.
- The system allows the implementation of suitable lighting profiles for each lamps on the basis of both traffic and weather conditions and complying with legislative limits and safety considerations.
- The experimental stage has demonstrated that the proposed system is capable to achieve a significant energy saving with respect to traditional street lamps systems.
- For further details about the project visit the web link www.smartlightingproject.it/new.

Acknowledgements

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